



MATHEMATICS

2014 HSC Course Assessment Task 4

August 13, 2014

General Instructions

- Working time – 55 minutes (plus 5 minutes reading time).
- Write using blue or black pen. Diagrams may be sketched in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt all questions.

This paper contains 4 questions.

- Commence each new question on a new page.
- Show all necessary working for every question.
- Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER:

BOOKLETS USED:

Class (please ✓)

☐

Mr Lin

☐

Mr Berry

☐

Mr Weiss

☐

Mr Lowe

☐

Mr Zuber

Question	1	2	3	4	Total
Marks	$\overline{6}$	$\overline{6}$	$\overline{9}$	$\overline{12}$	$\overline{33}$

Question 1 (6 marks)

An object, initially at rest at the origin, moves in a straight line with velocity $v \text{ ms}^{-1}$ so that $v = 4t(5 - t)$ where t is the time elapsed in seconds.

Find:

- (i) The acceleration of the object at the third second. 2
- (ii) An expression for the displacement (x) of the particle in terms of t . 2
- (iii) The position for the particle where it again comes to rest. 2

Question 2 (6 marks)

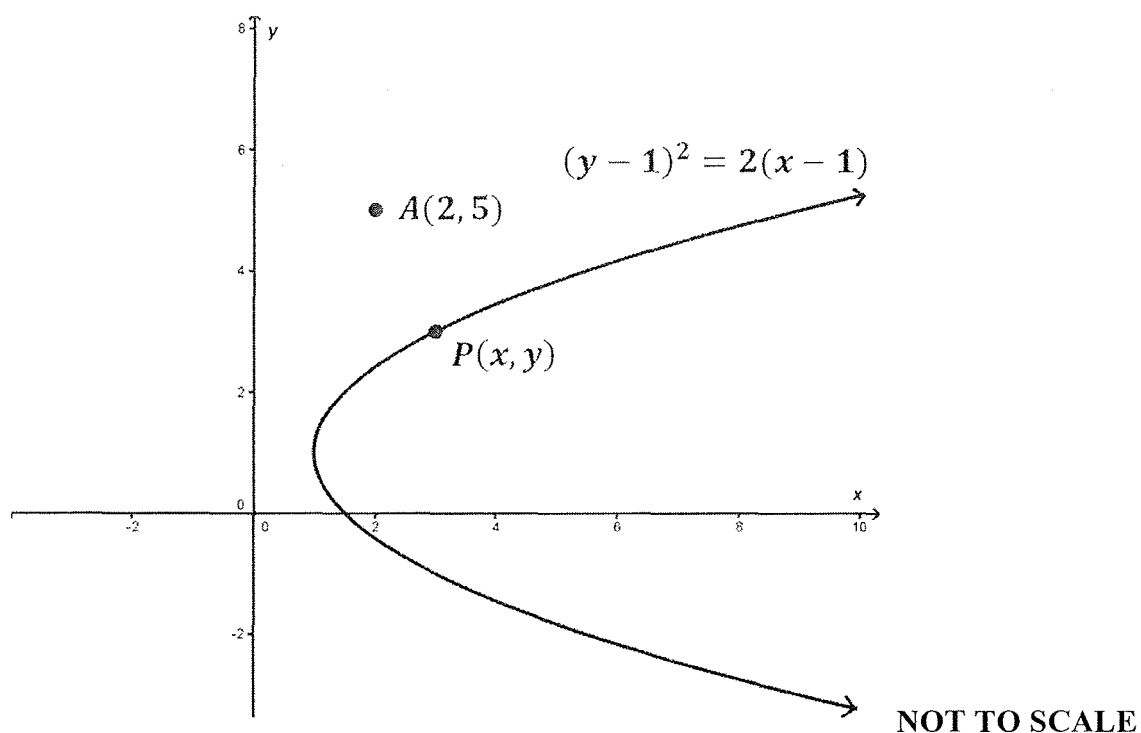
There are three cards marked X, Y, and Z and three envelopes also marked X, Y, and Z respectively. One card is randomly placed in each envelope.

What is the probability that:

- (i) All the cards are put into an envelope with the same letter as the card? 1
- (ii) No cards are put into envelopes with the same letter as the card? 1
- (iii) Exactly one card is placed into an envelope with the same letter as the card? 2
- (iv) At least one of the cards is put into an envelope with the same letter as the card? 2

Examination continues on the next page.

Question 3 (9 marks)



The diagram above shows the graph of the parabola $(y - 1)^2 = 2(x - 1)$. The point $A(2, 5)$ is outside the parabola while the point $P(x, y)$ is on the parabola.

- (i) If D is the distance between the points A and P , show that **3**

$$D^2 = \left[\frac{1}{2}(y - 1)^2 - 1 \right]^2 + (y - 5)^2$$

- (ii) Find $\frac{d(D^2)}{dy}$ and show that the value of D in the equation in part (i) is a **4**

minimum when $y = 3$.

- (iii) Hence or otherwise find the distance between A and P and find the coordinate of the point P **2**

Examination continues on the next page.

Question 4 (12 marks)

A fund is established to provide birthday presents for Bill. \$10 000 is placed in the fund when Bill was born. It is decided that \$450 will be drawn from the fund each year to purchase his birthday presents. The money in the fund is invested 3% p.a. compounded annually with the interest paid into the fund before each of his birthdays.

- (i) Show the fund contains \$9695.50 after Bill's second birthday. 2
- (ii) If A is the amount in dollars remaining in the fund after n^{th} birthday, prove that $A_n = 5000(3 - 1.03^n)$. 3
- (iii) Find the amount of money in the fund after Bill's 25th birthday. Give your answer correct to the nearest dollar. 1
- (iv) Find the maximum number of birthday presents that can be financed using this fund. 2
- (v) For the fund described above, after Bill's first birthday his parents decided to increase the amount of money withdrawn for each birthday by 2% each year.
 - (a) Show that the amount remaining after Bill's 2nd birthday is \$9686.50 2
 - (b) Find the amount remaining in the fund after Bill's 25th birthday. 2
Give your answer correct to the nearest dollar.

End of examination.

Question 1

$$V = 4t(5-t)$$

$$= 20t - 4t^2$$

$$a) \frac{dv}{dt} = 20 - 8t$$

$$\text{when } t=3$$

$$\frac{dv}{dt} = 20 - 8 \times 3$$

$$= 20 - 24$$

$$= -4$$

$$b) x = \int 20t - 4t^2 dt$$

$$= 10t^2 - \frac{4t^3}{3} + C$$

$$\text{when } t=0 \quad x=0$$

$$\therefore C=0$$

$$\therefore x = 10t^2 - \frac{4t^3}{3}$$

$$c) \text{ comes to rest: } v=0$$

$$\text{at } t=5$$

$$\text{when } t=5, \quad x = 10(5^2) - \frac{4}{3}5^3$$

$$= 2 \times 5^3 - \frac{4}{3} \times 5^3$$

$$= \frac{2}{3} \times 5^3 = -\frac{250}{3} \text{ m.}$$

Question 2

a) $P(\text{all cards go into correct envelopes})$

$$= P(ABC)$$

$$= \frac{1}{3} \times \frac{1}{2} \times 1$$

$$= \frac{1}{6}$$

b) $P(\text{no cards go into correct envelopes})$

$$= P(\bar{A}\bar{B}\bar{C})$$

$$= \frac{2}{3} \times \frac{1}{2}$$

$$= \frac{1}{3}$$

c) $P(\text{exactly one card})$

$$= P(A\bar{B}\bar{C} \text{ or } B\bar{A}\bar{C} \text{ or } C\bar{A}\bar{B})$$

$$= \frac{1}{3} \times \frac{1}{2} \times 1 + \frac{1}{3} \times \frac{1}{2} \times 1 + \frac{1}{3} \times \frac{1}{2} \times 1$$

$$= \frac{1}{2}$$

d) $P(\text{at least 1 card})$

Method 1

$$P(1 - P(\text{no cards correct}))$$

$$= 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

Method 2

$$P(1 \text{ card}) + P(\text{all cards})$$

$$= \frac{1}{2} + \frac{1}{6}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

Question 3

$$D^2 = (x-2)^2 + (y-5)^2$$

$$(y-1)^2 = 2(x-1)$$

$$(x-1) = \frac{(y-1)^2}{2}$$

$$\therefore x = \frac{(y-1)^2}{2} + 1$$

$$\therefore D^2 = \left[\frac{(y-1)^2}{2} + 1 - 2 \right]^2 + (y-5)^2$$

$$= \left[\frac{(y-1)^2}{2} - 1 \right]^2 + (y-5)^2$$

$$\therefore \frac{dD^2}{dy} = 2 \left[\frac{(y-1)^2}{2} - 1 \right] \cdot (y-1) + 2(y-5)$$

$$= (y-1) \left[\frac{(y-1)^2}{2} - 1 \right] + 2(y-5)$$

$$= (y-1)^3 - \frac{2}{2}(y-1) + 2(y-5)$$

$$= (y-1)^3 - 8$$

$$= [(y-1) - 2] [(y-1)^2 + 2(y-1) + 4]$$

$$= (y-3) [y^2 - 2y + 1 + 2y - 2 + 4]$$

$$= (y-3) [y^2 + 3]$$

$$y^3 + 3y - 3y^2 - 9$$

$$\text{Let } \frac{dD^2}{dy} = 0 \Rightarrow (y-3)(y^2+3) = 0$$

$$y = 3$$

$$\frac{d^2(D^2)}{dy^2} = 3y^2 - 6y + 3$$

$$\text{when } y = 3$$

$$\frac{d^2(D^2)}{dy^2} = 27 - 18 + 3$$

$$= 12$$

$$> 0$$

$\therefore$ minimum at $y = 3$

c) when $y = 3$.

$$D^2 = \left[\frac{1}{2} (3-1)^2 - 1 \right]^2 + (3-5)^2$$

$$= [1^2] + (-2)^2$$

$$= 5$$

$$\therefore D = \sqrt{5}, \quad D > 0.$$

$$D > 3$$

Question 4

1. a) $A_1 = \$10000 \times 1.03 - 450$

$$\begin{aligned} A_2 &= (\$10000 \times 1.03 - 450) \times 1.03 - 450 \\ &= \$10000 \times 1.03^2 - 450(1 + 1.03) \\ &= \$9695.50 \end{aligned}$$

b) $A_3 = (\$10000 \times 1.03^2 - 450 \times 1.03 - 450) \times 1.03 - 450$

$$= 10000 \times 1.03^3 - 450 \times 1.03^2 - 450 \times 1.03 - 450$$

$$= 10000 \times 1.03^3 - 450(1 + 1.03 + 1.03^2)$$

$$\therefore A_n = 10000 \times 1.03^n - 450 \underbrace{(1 + 1.03 + 1.03^2 + \dots + 1.03^{n-1})}$$

GP: $a=1$ ~~no~~ $r=1.03$

$$= 10000 \times 1.03^n - 450 \left(\frac{1.03^n - 1}{0.03} \right)$$

$$= 10000 \times 1.03^n - 15000(1.03^n - 1)$$

$$= 10000 \times 1.03^n - 15000 \times 1.03^n + 15000$$

$$= ~~8500 \times 1.03^n~~ 15000 - 5000 \times 1.03^n$$

$$= 5000(3 - 1.03^n)$$

c) when $n=25$

$$A_{25} = 5000(3 - 1.03^{25})$$

$$= \$4531 \text{ to the nearest dollar}$$

d) $A_n = 0 \Rightarrow 5000(3 - 1.03^n) = 0$

$$3 - 1.03^n = 0$$

$$1.03^n = 3$$

$$n = \frac{\log_e 3}{\log_e 1.03}$$

$$= 37.17$$